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Towards Accurate and Interpretable Competency-Based Assessment: Enhancing Clinical Competency Assessment through Multimodal AI and Anomaly Detection

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Abstract

Artificial Intelligence (AI) is reshaping medical education, particularly in the domain of competency-based assessment, where current methods remain subjective and resource-intensive. We introduce a multimodal AI framework that integrates video, audio, and patient monitor data to provide objective and interpretable competency assessments. Using 90 anesthesia residents, we established "ideal" performance benchmarks and trained an anomaly detection model (MEMTO) to quantify deviations from these benchmarks. Competency scores derived from these deviations showed strong alignment with expert ratings (Spearman's $\rho = 0.75$; ICC $= 0.75$) and demonstrated high ranking precision (Relative L_2 −distance = 0.124). SHAP analysis revealed that communication and eye contact with the patient monitor are key drivers of variability. By linking AI-assisted anomaly detection with interpretable feedback, our framework addresses critical challenges of fairness, reliability, and transparency in simulation-based education. This work provides actionable evidence for integrating AI into medical training and advancing scalable, equitable evaluation of competence.

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Introduction

Assessing the competencies of healthcare professionals is essential for ensuring patient safety and delivering high-quality care[1-3]. Traditional skill assessment methods rely heavily on subjective evaluations and standardized tests, which can be limited in capturing practitioners' nuanced performance in real-world scenarios[1,3-6].

Simulation-based assessment (SBA) offers a powerful complement, enabling the evaluation of clinical competency within a safe, controlled, and highly authentic environment[6]. By recreating complex patient care scenarios, SBA can assess domains that are often overlooked by traditional skill assessment methods, such as managing acute physiological deterioration, prioritizing interventions, and adapting to evolving clinical contexts[7,8]. Nonetheless, performance evaluation in SBA is inherently challenging: expert evaluators must meticulously observe and judge each action and decision point[7,9], and compare them to a "gold standard" clinical practice[7,9,10]. This makes the evaluation process resource-intensive, underscoring the need for robust SBA frameworks that can compare physicians' skills with clear performance criteria to ensure objectivity and validity. These challenges echo long-standing issues in SBA, which include standard setting for high-stakes decisions, balancing formative vs. summative use, and the need to sample authentic performance across cases[7,9].

Advances in artificial intelligence (AI) have enabled automated skill assessment systems capable of decomposing motor skills into chronological actions and evaluating maneuver execution[4,11-17]. Yet, most approaches rely on coarse categorical outcomes (e.g., high vs. low proficiency), which can obscure the complexity of context-dependent performance and constrain formative feedback[18-20]. Furthermore, automated skill assessment research has primarily targeted procedural tasks recorded on video[4,11,21-23]. This focus, useful for standardization and reproducibility, also constitutes a limitation, because these tasks emphasize technically uniform, prescribed movements[24,25], and may under-represent context-sensitive and adaptive behavior seen in real clinical settings. "Skill," per se, extends beyond mere execution, and as a direct result, the granularity level of current automated skill assessment frameworks often fails to capture experts' detailed performance.

More recently, Kiyasseh et al.[26] and Igaki et al.[14] have explored the potential of automated skill assessment on intraoperative videos, demonstrating feasibility while also underscoring issues of generalizability and fairness that motivate context-aware and potentially multimodal approaches. Yet, these works are procedure-specific and predominantly video-based; they do not explicitly integrate multimodal information (e.g., human-object interaction, communication), which may enhance the assessment process. A complementary body of work shows that multimodal signals can enrich assessment beyond execution alone. For example, utilizing video and kinematic data from JIGSAWS[27] (JHU-ISI Gesture and Skill Assessment Working Set) for skill assessment[28], eye-tracking as an indicator of

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situational awareness in nursing education[29,30], and EEG/fNIRS measures that index cognitive workload and pre-execution processes (e.g., motor planning)[31-33].

We propose a paradigm shift toward multivariate time series analysis, integrating multiple data streams to generate a high-resolution, dynamic view of clinical performance[34]. Our framework extracts complementary signals from simulation recordings, each hypothesized to be a strong indicator of clinical competency:

1. **Human-object Interaction.** Patterns of engagement with medical tools and equipment.
2. **Movement and Spatial Positioning.** Body movements and location in simulation space.
3. **Verbal Expression.** Communication skills and indicators of cognitive state.

This multimodal representation supports richer and more comprehensive assessments that reveal subtle distinctions that may be imperceptible in other skill assessment methods. For this research, we are particularly interested in utilizing time series of expert-level execution to gain a fine-grained benchmark for competency comparisons.

Building on this foundation, multivariate anomaly detection identifies deviations from expected patterns across modalities[35,36]. Anomaly detection models define "normal" patterns as time series sequences that exhibit smoothness, predictability, and consistent local temporal structure. Deviations from these expectations are quantified as "anomaly scores", with higher scores indicating a greater likelihood of abnormality. In clinical skill assessment, this is advantageous when integrating diverse signals such as ours. By modeling the extracted time series of "ideal" (expert-level) execution as the "normal" baseline, deviations can be detected and objectively ranked according to their magnitude.

For anomaly detection, we employ Song et al.'s memory-guided Transformer for multivariate time series (MEMTO)[37], which maintains an adaptive memory of standard patterns and incrementally updates these representations as new data arrive. Once expert-level patterns are established, our model can detect and quantify deviations from the "ideal" baseline, providing a robust and objective ranking of clinical competency.

Lastly, consistent with recent simulation-education work on AI explanations, we also incorporated explainable AI analysis, which provided explanations as role-appropriate aids for instructor-led debriefs[38,39].

Recent work in automated skill assessment has begun to incorporate interpretable models and time-series analysis, including SHAP-based explanations for surgical performance[40-42]. However, these frameworks typically rely on single modalities, temporally aggregated features, or supervised models trained directly on rubric scores, and thus do not construct an explicit, expert-anchored baseline for behavior over time. In contrast, our framework fuses four complementary time-varying signals (monitor-directed gaze, spatial position, speech activity, and patient physiological trends) into a unified

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representation of medical emergency behavior, uses MEMTO to learn prototypical expert patterns and quantify deviations from those patterns at high temporal resolution, and then calibrates these anomaly-based scores to examiner rankings while providing phase-specific SHAP analyses of the underlying behavioral and physiological drivers.

In this study, we introduce a multimodal framework that integrates audio, video, and patient monitor data to move beyond surface-level action scoring and toward a richer evaluation of clinical performance in complex patient care scenarios. By applying multivariate anomaly detection, we model expert performance as the "gold standard" and translate deviations into an intuitive competency score, while explainability analysis highlights which inputs most influenced the model's anomaly score in our dataset; we treat these explanations as descriptive aids for debriefing rather than causal claims[39]. Together, this approach provides a practical, objective, and interpretable pathway for AI-assisted training and evaluation systems, laying the groundwork for next-generation systems that offer more comprehensive, clinically meaningful, and actionable feedback on clinical competency.

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Results

We recruited 90 (64.44% male) anesthesia residents from 17 hospitals, most of whom were in their 5th year of training (range: 4 - 10). On average, participants completed 5.80 ± 7.86 simulation examinations during residency (range: 0 - 60). A complete description of the participants is presented in Table 1.

For our data collection, each simulation was delivered in four predefined stages: Initial Condition (baseline assessment under stable vitals), Extreme Deterioration (rapid physiologic decline and alarm escalation requiring recognition and preparation), ACLS Management (active resuscitation per institutional protocols, e.g., chest compressions/airway/defibrillation/medications), and Back to Life (post-resuscitation stabilization and handoff). Each simulation was ranked by a board-certified anesthesiologist using a standardized scenario script and rating guidelines. The examiner ranked the overall performance on a scale from 1 to 5, where '1' indicates poor performance.

Table 1: Demographic summary. The averaged values are presented as mean ± standard deviation.

Total Number of Participants = 90		**Year of Residency**			**Simulation exams during residency**		
Sex	**Count**	**Year**	**Count**	**Mean±STD**	**Range**	**Count**	**Mean±STD**
Male	58 (64.44%)	4	14	5.48 ± 1.32	0 – 3	25	6.21 ± 9.31
		5	20		4 – 6	22	
		6	16		7 – 9	3	
		7+	8		10+	8	
Female	32 (35.56%)	4	12	5.09 ± 1.32	0 – 3	14	5.06 ± 4.17
		5	12		4 – 6	10	
		6	3		7 – 9	2	
		7+	5		10+	6	

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Competency Assessment

Figure 1 presents the distribution of anomaly scores per population based on the MEMTO model trained on the Rank 5 population. Table 2 presents a coarser analysis of the anomaly scores.

Figure 1: MEMTO anomaly scores magnitude and frequency across ranks.
a) Normalized histograms (density) of MEMTO anomaly scores (range 0–1) for each rank group (Rank 1–Rank 5), showing a shift toward lower anomaly scores with increasing performance rank. The number of detected anomalies and the average anomaly score for each rank group are shown in the inset boxes (Rank 1: blue bars; Rank 2: orange bars; Rank 3: green bars; Rank 4: red bars; Rank 5: purple bars). **b)** Boxplots of MEMTO anomaly scores stratified by performance rank group, demonstrating decreasing anomaly scores with higher ratings.

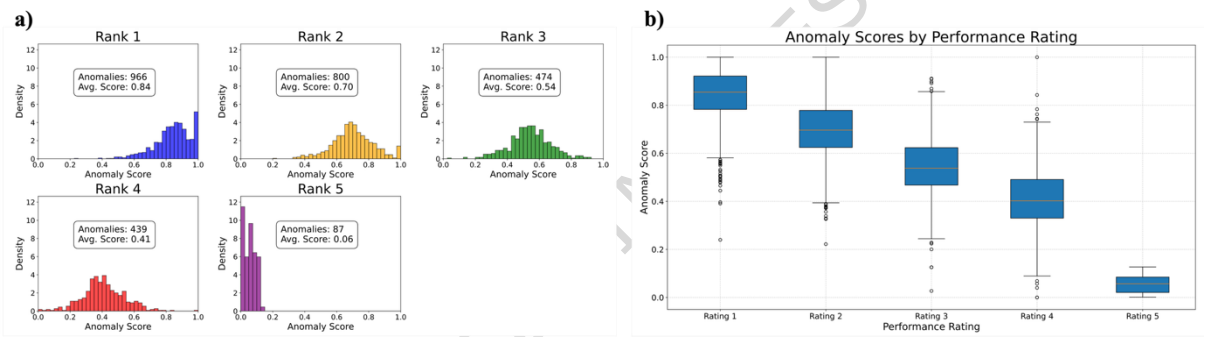

Table 2: Average number and value of anomalies per time series across performance rankings.

Time Series	Metric	Rank 1	Rank 2	Rank 3	Rank 4
Gaze	Avg. Anomalies	19.64 ± 4.07	20.33 ± 2.67	14.30 ± 5.82	10.83 ± 1.55
	Avg. Score	0.84 ± 0.12	0.70 ± 0.15	0.54 ± 0.10	0.41 ± 0.09
Movement	Avg. Anomalies	12.09 ± 3.98	10.24 ± 2.31	10.15 ± 3.47	6.08 ± 1.85
	Avg. Score	0.72 ± 0.10	0.58 ± 0.11	0.44 ± 0.09	0.35 ± 0.07
Speech	Avg. Anomalies	14.05 ± 1.46	13.68 ± 1.04	9.47 ± 1.82	9.54 ± 1.29
	Avg. Score	0.88 ± 0.14	0.76 ± 0.13	0.61 ± 0.11	0.52 ± 0.10

Gaze – Physician's visual examination of the patient monitor
Movement – Physician location in the simulation area
Speech – Physician's verbal activity

Table 3 presents the effect of incrementally adding time series modalities on MEMTO model performance. We trained the model using the best-performing participants (Rank 5) as the reference population, as their execution patterns were assumed to represent the "ideal" clinical behaviors. For comparison, we also trained MEMTO using the worst-performing participants (Rank 1) to examine how learning from suboptimal performance affects model accuracy. These anchors correspond to two

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conceptually different ways of defining 'normal' behavior: similarity to high-quality expert performance versus similarity to systematically weak performance. Intermediate ranks produce performance between these two extremes, so we focus on Rank 1 and Rank 5 to illustrate how anchoring the model on expert versus poor patterns affects alignment with examiner ratings.

Table 3: Ablation study on the different time series for models trained on Rank 1 and Rank 5 data. The reported values are the overall (resident-level) estimates.

	Rank 1 – Worst Performance			**Rank 5 – Best Performance**		
Time Series	$\rho(\uparrow)$	$R-L_2(\downarrow)$	$ICC(1,k)(\uparrow)$	$\rho(\uparrow)$	$R-L_2(\downarrow)$	$ICC(1,k)(\uparrow)$
Gaze	0.011	0.932	0.143	0.286	0.795	0.157
Gaze + Movement	0.181	0.723	0.152	0.634	0.401	0.466
Gaze + Movement + Speech	0.389	0.701	0.373	0.752	0.129	0.703
Gaze + Movement + Speech + Vitals	0.423	0.659	0.503	0.776	0.124	0.748

ρ = Spearman's rank correlation coefficient; ICC(1,k) = Intraclass Correlation Coefficient, one-way random effects model, absolute agreement, and multiple raters. Unless mentioned otherwise, all correlation tests are two-sided with α=0.05.

Gaze – Physician's visual examination of the patient monitor; Movement – Physician's location in the simulation area; Speech – Physician's verbal activity; Vitals – Physiological measurements from the patient monitor.

Ninety-five percent confidence intervals, obtained via bootstrap over residents, indicated that the multimodal approach achieved markedly higher agreement with examiner rankings than single modality (e.g., Spearman's $\rho_{multi} = 0.776, 95\%\ \text{CI}\ 0.707 - 0.813$ vs. $\rho_{gaze} = 0.286, 95\%\ \text{CI}\ 0.169 - 0.325$). The bootstrapped difference across all evaluation metrics excluded zero, and a paired Wilcoxon signed-rank test on per-resident absolute errors confirmed that multimodal predictions were significantly closer to examiner scores than single and dual modality predictions. Table S2 in the Supplementary present a complete overview of the ablation study with the confidence intervals, and Table S3 contains the results for the statistical analysis.

Table S4 and Figure S1 in the Supplementary depict examples of anomaly intervals detected by the MEMTO model, respectively. Figure S2 in the Appendix presents the correlation analysis between the predicted and human evaluator competence scores.

Explainability - SHAP Analysis

Figure 2 displays the mean absolute SHAP values across simulation phases, highlighting which features most strongly influenced MEMTO's anomaly score predictions. In the ***Initial Condition***, verbal activity was the most influential indicator, suggesting that early communication patterns differentiated

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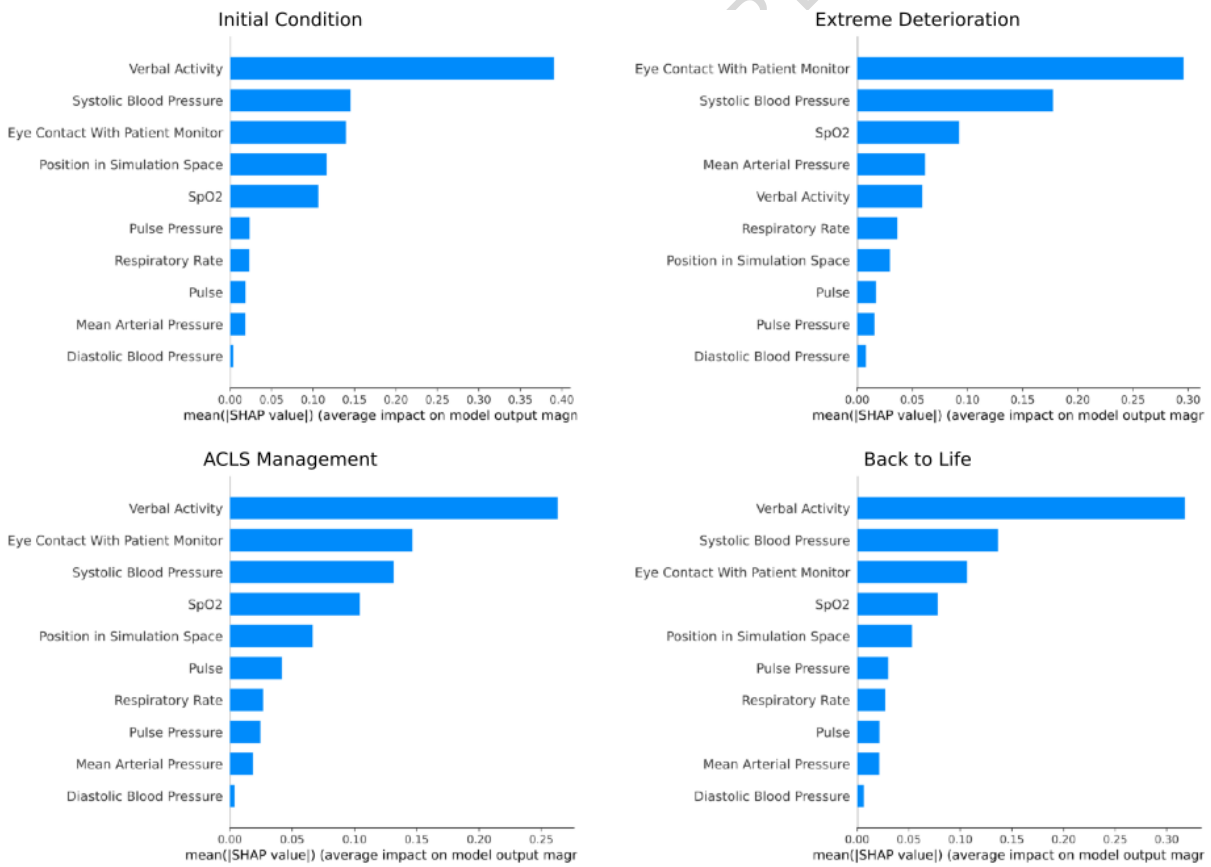
performance. During ***Extreme Deterioration***, eye contact with the patient monitor became the dominant feature, underscoring the critical role of sustained visual attention in crisis recognition. In the ***ACLS Management*** phase, verbal activity again ranked highest, together with eye contact and systolic blood pressure, reflecting the importance of coordinated communication and physiological monitoring during resuscitation. In the ***Back to Life*** phase, verbal activity remained the leading feature, while systolic blood pressure and SpO_2 gained prominence as markers of stabilization. Overall, the analysis shows that MEMTO's predictions relied most heavily on verbal (communication) and nonverbal behaviors (gaze, positioning), with vital signs providing complementary but phase-dependent contributions.

Figure 2. SHAP feature importance across simulation phases. Bars show the mean absolute SHAP value for each input time series, summarizing its average contribution to MEMTO's predicted anomaly score (larger values indicate greater average influence on the model output). Features are ordered from highest to lowest value.



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Discussion

We developed and validated a multimodal AI-based framework for simulation-based competency assessment that integrates gaze, movement, speech, and patient monitor signals into interpretable time series representations. By anchoring anomaly detection to expert-level performance patterns, residents with lower examiner-assigned rankings consistently exhibited more anomaly segments with higher anomaly scores, supporting the framework's ability to capture clinically meaningful deviations from expert practice. Importantly, the ablation analysis underscored the value of integrating multiple behavioral and physiological signals. When trained on expert (Rank 5) data, MEMTO achieved a Spearman correlation of $\rho = 0.78$ and ICC of 0.75 with gaze, movement, speech, and vitals combined, compared with $\rho = 0.29$ and ICC = 0.16 when using gaze alone. These findings indicate that, in this crisis-management setting, integrating behavioral and physiological signals captures complementary aspects of competency beyond any single channel. While single-modality approaches can perform well in some contexts, our results suggest added value for multimodal fusion when the competency construct spans communication, attention, and physiologic monitoring. Conversely, training on poor performance patterns (Rank 1) produced weaker alignment with human judgment, reinforcing the need to anchor benchmarks in expert-level execution.

Exploratory SHAP analysis suggested that, beyond quantifying deviations, our framework can indicate which specific modalities and phases were most informative for the model's decisions. In our cohort, speech activity often emerged as a dominant factor across phases, consistent with its role in team coordination and cognitive processing. Eye interaction with the patient monitor tends to contribute the most during crisis escalation, aligning with the clinical need for continuous situation awareness. Physiological signals, particularly systolic blood pressure and SpO_2, gained prominence during resuscitation and recovery, underscoring the importance of interpreting patient data during these phases. These patterns are descriptive, model- and background-dependent, and should be interpreted with caution, as they do not establish causality or demonstrate educational benefit. We view SHAP primarily as an aid to contextualize which inputs influenced a segment's anomaly score and to help structure instructor-led debriefs, not as a replacement for expert judgment; evaluating actionability and impact on learning will require dedicated user studies.

Together, these findings suggest that SHAP-based interpretability could support more actionable, phase-specific feedback than opaque numerical scores and may help structure instructor-led debriefs.

Our framework extends prior AI-assisted assessment beyond standardized procedural tasks. In surgery, many studies estimate rubric-based scores (e.g., OSATFS)[4,11,12,14,15], but these advances are largely limited to procedural settings and predominantly video-only data. Outside of surgery, automated

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assessment remains less mature and often execution-centric, with coarser performance grouping[16]. Accordingly, we model multimodal, non-procedural behaviors and map them to a continuous competency scale, answering the longstanding calls for more clinically authentic, comprehensive, and competency-based assessments. Unlike traditional examiner-based scoring, which is time-consuming and prone to variability[7,9,10], our approach offers a scalable, objective, and transparent alternative. In doing so, it aligns with current trends in medical education that emphasize fairness, reliability, and feedback-driven learning, while also advancing patient safety–driven training standards[3,4,11,15,16].

Our framework is intended as a criterion-referenced, human-in-the-loop complement to examiner judgment, not a replacement. Because inputs include audiovisual data, privacy and consent (IRB approval, secure storage, access controls, limited retention) are essential, and derived time-series views should be preferred over raw media for routine review. Although we exclude demographic variables, proxy bias remains possible. In exploratory post hoc analyses, we did not observe systematic differences in model performance across male and female residents. Yet, we recommend subgroup reporting, periodic fairness audits (e.g., by training year/cohort/site), and transparency when gaps are detected. In our setting, the Rank 5 reference group is drawn from residents training at 17 hospitals under a shared national board simulation curriculum, thereby approximating a national normative standard rather than the practice patterns of a single institution. Nevertheless, we view this benchmark as local to this program and healthcare system and do not assume that it represents a universal notion of "expert" performance across all institutions or countries. Responsible deployment requires site-specific re-benchmarking and recalibration, and future work should explicitly evaluate cross-institutional generalizability by testing the framework on external cohorts, constructing multi-site expert benchmarks, and quantifying the recalibration required when the model is transported to new simulation centers. SHAP explanations are post-hoc and baseline-dependent and should be used to contextualize model outputs during instructor-led debriefs rather than to automate high-stakes decisions. Future work should incorporate user-centered evaluation, rating the usefulness and trustworthiness of explanations, and, ideally, controlled comparisons of debriefs with versus without SHAP support, to determine whether these explanations measurably improve feedback quality or learning outcomes.

Several limitations should be noted, which also suggest directions for future work. First, our study was restricted to 90 anesthesiology residents in high-fidelity simulation environments, which may limit generalizability to other specialties or real-world clinical practice. Second, while anchoring anomaly detection to expert benchmarks enhances validity, it assumes that expert performance is both homogeneous and universally ideal, potentially overlooking safe but alternative strategies. Third, our operational definition of "expert" relied on the highest-performing residents (Rank 5) as the reference group. This approach enables fairer peer-to-peer comparisons on a 1–5 scale; however, it may

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underestimate the performance gap relative to an experienced, board-certified specialist. Hybrid benchmarking, which combines resident and certified expert data, could provide a more comprehensive standard while preserving interpretability. From an assessment-design perspective, our implementation also leaves the underlying construct only partially explicit. Traditional frameworks such as evidence-centered design[43] and construct modeling[44] distinguish between defining the latent construct, specifying how task performances provide observable evidence for that construct, and accumulating that evidence into scores. In our framework, these layers are partly entangled: MEMTO simultaneously learns which multimodal patterns are typical of our expert benchmark and how deviations from those patterns should be combined into an anomaly-driven competency score. As a result, the construct we operationalize is largely instantiated empirically as similarity to the observed behavior of Rank 5 residents in this program, even though scores are used in a criterion-referenced manner, and the mapping from individual anomalies to specific latent abilities is only partially knowable, even with post hoc explanations. Future work should therefore integrate more explicit construct maps and evidence models, using MEMTO-derived anomalies as one layer of evidence within a broader measurement model. Finally, although SHAP analysis highlighted the influence of different modalities on anomaly scores, the anomalies themselves remain somewhat abstract. For example, during the *Extreme Deterioration* phase, eye contact with the patient monitor contributed the most to the anomaly score, yet our framework does not specify what constitutes optimal behavior (e.g., frequency, duration, or distribution of monitor checks). Future work should therefore aim to translate anomalies into more fine-grained behavioral descriptors that trainees can apply to improve their performance.

In conclusion, this study presents a paradigm shift in simulation-based assessment, moving from subjective, binary evaluations toward multimodal, anomaly-driven competency scoring. By quantifying deviations from expert performance and revealing their underlying behavioral and physiological drivers, our framework demonstrates how AI can provide objective, interpretable, and educationally meaningful assessments. This work highlights the potential for AI to support competency-based medical education, ultimately enhancing both training and patient safety.

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Methods

Medical Simulations Dataset

The Israel Society of Anesthesiologists conducts a biannual, two-day workshop designed to prepare final-year anesthesia residents for their national board certification. The workshop was designed to closely replicate the format and expectations of the official exam, with a specific focus on the SBA component.

Each simulation involved two research team members, who portrayed the roles of a nurse and a medical intern, while a board-certified anesthesiologist evaluated the resident's performance. All examiners had prior experience conducting SBA for the national board certification examination in anesthesiology. To reduce variability, each examiner was assigned a single crisis scenario and used a standardized scenario script and accompanying rating instructions. These instructions directed examiners to integrate multiple observable domains into a global judgment of performance. At the end of the evaluation, the examiner provided an overall performance rank on a scale from 1 to 5, with Rank 1 indicating substandard performance and Rank 5 indicating exemplary performance. The final rank, therefore, reflected a holistic judgment across these domains rather than a single technical maneuver and served as our ground truth for the framework (see Table S1 in the Supplementary).

The simulations were delivered using a 'Laerdal' full-body high-fidelity manikin in a fully equipped simulation suite (Figure 3-A). Video coverage of the simulation was captured from two angles:

1. Overview of the simulation area. Provides information regarding the subject's movement and location (Figure 3-A)
2. Patient monitor point-of-view. Capturing the subject's interaction with the monitor. Additionally, the patient monitor display was also recorded (Figure 1-B).

All participants, including the research team, wore wireless Lavalier microphones to ensure high-quality audio capture. Videos were recorded at 30 FPS with 1080p resolution and the MPEG-4 codec; audio was captured at 44.1 KHz with 16-bit depth. All streams were time-aligned using StreamPix digital video recording software (version 9.3.1.0), provided by NorPix Inc (https://www.norpix.com/).

The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. All participants gave written informed consent to take part in the simulations and for their data to be analyzed for research purposes. The Rambam Health Care Campus Institutional Review Board approved this study (approval number: 0482-20-RMB).

Seven clinical simulation scenarios (Table S1 in the Supplementary) were developed by an experienced anesthesiologist in collaboration with a medical simulation specialist. To ensure clinical authenticity and alignment with board-exam standards, each scenario was divided into four chronologically distinct phases: (1) ***Initial Condition*** – the participant assesses the patient's status by

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examining the manikin and communicating with the nurse; (2) ***Extreme Deterioration*** – the patient experiences a rapid and severe decline in physiological status (e.g., hypotension, tachycardia), requiring immediate crisis management; (3) ***ACLS Management*** – the resident executes urgent interventions with speed and accuracy, demonstrating mastery of Advanced Cardiac Life Support protocols; and (4) ***Back to Life*** – the participant updates the team, coordinates ongoing care, and transitions to post-crisis management. Each phase is bounded by clearly defined start and end timestamps, marked by observable cues on the patient monitor and, in some cases, on the manikin.

Construction of Multivariate Time Series

We implemented an end-to-end pipeline for constructing multivariate time series from video recordings of a medical simulation. This process involved designing and adapting various computer vision models, as well as developing custom preprocessing steps to ensure high temporal alignment and data quality. Figure 3 provides an overview of the proposed framework, in which the primary data sources are the subject's actions and the patient's physiological monitor. Figure 4 provides an overview of the complete pipeline, from raw recordings and time series construction, through MEMTO-based anomaly scoring, to the final competency scores and SHAP-based explanations.

To capture the behavioral dynamics of the clinician during the simulation, we developed and applied a series of processing stages that transformed raw audiovisual recordings into high-resolution, synchronized time series.

- Physician-Monitor Interaction. We adopted a webcam-on-monitor approach, in which a commercial webcam was mounted on the top edge of the patient monitor screen. Building on our previously validated end-to-end gaze-target pipeline[45], each frame is processed with face detection, tracking, and spatiotemporal gaze target estimation to determine whether the physician's gaze intersects the region of interest. Frames in which the estimated gaze vector intersected this region are labeled as "looking at the monitor" and all other frames are labeled as "not looking at the monitor", yielding a binary time series of monitor interaction (Figure 3-C). We extended our previous system to handle multiple faces in the scene, enabling precise isolation of the subject's gaze toward the patient monitor. This was achieved by replacing the original face detector with a pre-trained YOLOv10[46]-based face detector (https://github.com/YapaLab/yolo-face, yolov10l-face.pt) and integrating the BoT-SORT algorithm[47] (https://github.com/NirAharon/BoT-SORT) for multi-person tracking with re-identification across occlusions to maintain subject identity over time. Both YOLOv10 and the BoT-SORT algorithm were used in inference mode, without any changes to their default parameters. The models' training & hyperparameter selection details are provided in their respective GitLab webpages.

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The gaze-target detector and tracking components are used in inference-only mode and were not retrained on this dataset, so no additional manual labels from these simulations were required.

- 3D Pose Estimation. We implemented a hybrid 3D pose tracking pipeline that fuses state-of-the-art detection, tracking, and 3D human reconstruction: (i) YOLOv10 for real-time person detection (https://github.com/ultralytics/ultralytics), providing accurate frame-level bounding boxes for all individuals; (ii) BoT-SORT for appearance-aware multi-person tracking with re-identification, maintaining stable identities across occlusions and motion; and (iii) TRAM[48] to recover frame-by-frame 3D human meshes and global, metric-scale trajectories in world coordinates from the recorded video (https://github.com/yufu-wang/tram). We used the fused outputs to obtain a continuous time series of only the resident and formed a 2-D standing-location time series by projecting the resident's root (pelvis) trajectory onto the ground plane and then expressing the result in a patient-centric coordinate system. This yields frame-aligned coordinates (x_t, y_t) relative to the patient (Figure 3-E). The models' training & hyperparameter selection details are provided in their respective GitLab webpages.

- Audio Processing. Although the resident's audio track was recorded on a separate channel, it contained speech overlap and ambient noise due to the close proximity of the simulation participants. To isolate the president's speech, we used the ESPnet-SE toolkit[49] (https://github.com/espnet/espnet) in inference-only mode with a 16 kHz single-channel pretrained DCCRN enhancement model[50] (no fine-tuning). The enhanced waveform was then analyzed with short-time log-energy features (25 milliseconds frame**,** 10 milliseconds hop), and we derived a binary speech-activity sequence using an adaptive energy threshold with hysteresis (minimum speech = 200 milliseconds**,** minimum silence = 200 milliseconds) with a brief morphological smoothing of 100 milliseconds to bridge micro-gaps within utterances. The resulting speech/no-speech decisions were aligned with the 30 FPS video timeline**,** marking a frame as active if any part of its 33.3-millisecond interval overlapped with detected speech. This constructed a resident-only speech-activity time series (Figure 3-D).

We extracted the patient's physiological signals directly from the monitor display using an optical character recognition model (DocTR[51]), adapting the pipeline to our monitor's fixed layout by defining region-of-interest (ROI) masks for each readout (heart rate, systolic/diastolic blood pressure, respiratory rate, SpO_2). For every video frame, DocTR produced text boxes and transcriptions; within each ROI, we retained digit-only strings, parsed them to numeric values, and discarded entries that violated simple physiological range checks (Figure 3-B). Because non-invasive blood pressure values appear

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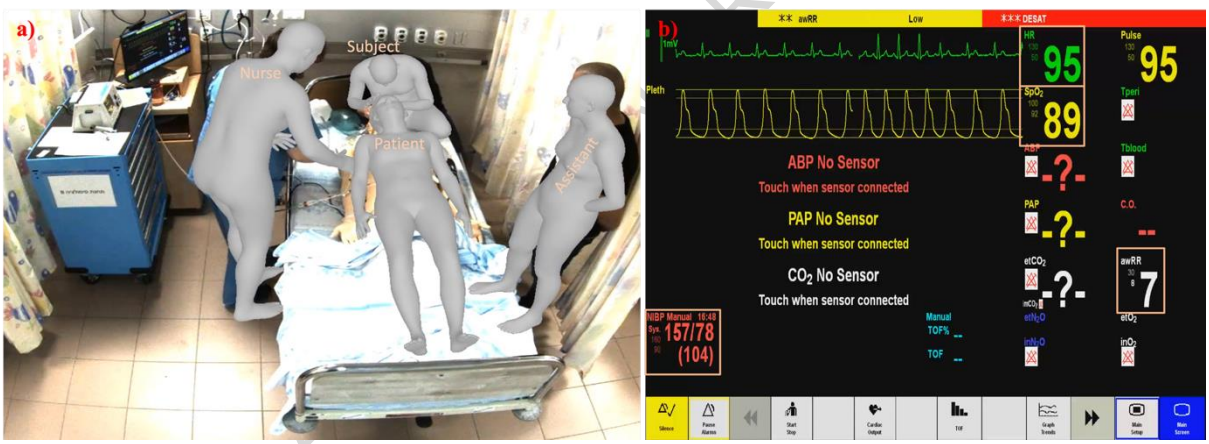
intermittently, we forward-filled between successive valid readings (with a capped hold time) and treated longer gaps as missing; short gaps in continuously displayed channels (heart rate, respiratory rate, SpO_2) were linearly interpolated to maintain continuity. From systolic and diastolic blood pressure, we derived pulse pressure (PP = S−D) and mean arterial pressure (MAP = (S+2D)/3). All series were time-aligned to the 30 FPS video timeline so that the monitor signals and the subject-action streams shared a common frame index.

Figure 3: Construction of the simulation time series.

a) Example frame showing three-dimensional (3D) body meshes overlaid on the simulation scene for each participant (nurse, subject, assistant) and the patient. The subject is manually identified, and the stationary patient serves as a spatial reference to ensure consistent tracking and alignment over time. **b)** Example patient-monitor screen capture used to extract time-varying vital signs. An orange rectangular region of interest highlights the monitor area containing numeric vital signs readouts used for downstream analysis (e.g., heart rate and peripheral oxygen saturation).

Data acquisition from subject actions and the patient monitor.



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Anomaly Criterion and Competency Score Calculation

Building on the constructed multivariate time series, we developed an in-house procedure to quantify deviations from expert-level ("ideal") performance and translate these deviations into an interpretable competency score (1-5) using MEMTO (code: https://github.com/gunny97/MEMTO). MEMTO is explicitly designed for multivariate time series anomaly detection, maintaining a memory of prototypical 'normal' patterns and providing competitive performance on high-dimensional benchmark datasets. These properties align with our need to detect subtle, temporally localized deviations across multiple behavioral and physiological channels, rather than relying on handcrafted anomaly rules.

Our competency evaluation approach was designed around the principle that expert-level performance can serve as an operational definition of the "ideal" execution pattern. We first train MEMTO only on Rank 5 (expert) simulations using a reconstruction-based objective, without using examiner ranks as

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targets, thereby establishing a fixed expert benchmark and then evaluating all other performances relative to this reference. This ensures that anomaly detection is anchored to a clinically meaningful competency standard.

This process comprises four main stages:

1. Model Training on Expert Reference Patterns. We assembled a reference dataset exclusively from the highest-performing residents (Rank 5), as rated by human experts.
   MEMTO couples a Transformer encoder with a deliberately weak, two-layer decoder (to reduce over-generalization) and an external gated memory. By training MEMTO only on these "expert" sequences, we allow the model to learn prototypical "normal" temporal patterns, i.e., to embed the notion of "ideal" time series patterns. To stabilize memory learning, we follow MEMTO's two-phase training: pretrain on reconstruction, then initialize memory items with K-means on encoder features before continuing training with gated updates. This yields an expert-anchored model of expected patterns across modalities and stages.
2. Anomaly Detection. We employed the expert-trained MEMTO to compare time series from other participants (lower-ranked) against the established "ideal" patterns. MEMTO yields per-timepoint anomaly scores by combining two signals: (i) the distance from the encoder query to its nearest memory item (larger for non-expert patterns because memory items encode normal prototypes), and (ii) the reconstruction error in the input space. Following the MEMTO paper, we normalize the distance using the softmax function and use it to weight the reconstruction error elementwise, thereby producing sharper separation between normal and abnormal windows. This allows MEMTO to identify deviations across modalities and flag moments within the time series that are inconsistent with expert-level patterns (i.e., anomalies). MEMTO was trained in the same reconstruction-based manner as described by Song et al. For each expert (Rank 5) simulation, the model minimizes a combination of an L_2 reconstruction loss between the input sub-series X_s and their reconstructed $\widehat{X_s}$, and an entropy regularization term on the memory attention weights. Specifically for our dataset, we used the following hyperparameters: $\lambda = 0.001$, 13 memory items, and optimized the model with AdamW.
3. Competency Score Calculation. Each detected anomaly is quantified and aggregated into an overall anomaly score, where higher values indicate greater deviation from the "ideal" patterns.
   The aggregated anomaly score is then monotonically calibrated to the examiners' 1–5 rubric (higher = more expert-like), so the predicted score remains transparent and clinically interpretable.
4. Model Validation. To ensure robustness, we perform a Leave-One-Out Cross-Validation (LOOCV) within the Rank 5 cohort: in each fold, we hold out one expert resident session for testing, while training on the remaining experts' time series; the model is expected to recognize the held-out sample

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as expert-like.

Evaluation Metrics

To ensure the accuracy and reliability of our methodology, we employ several evaluation metrics:

1. Spearman's rank correlation (ρ). This measures how well our model's competency scores align with those of expert evaluators. A higher correlation indicates better agreement.
2. Relative L_2 distance ($R - L_2$). This metric measures the discrepancy between our model's predictions and those of the expert evaluators. A lower value indicates better performance.
3. Intraclass correlation. This measures the consistency between the expert evaluators and our model's scores, providing an additional layer of validation. We calculated the ICC score using a one-way random effects model, absolute agreement, and multiple raters, denoted as ICC(1, k). A higher value indicates greater agreement.

To quantify uncertainty, we obtained 95% confidence intervals for all evaluation metrics via nonparametric bootstrapping over residents (2000 resamples). For key comparisons between modality combinations, we assessed statistical significance in two ways: (i) by bootstrapping the difference in the evaluation metrics and inspecting whether the 95% confidence interval excluded zero, and (ii) by applying paired Wilcoxon signed-rank tests to compare per-resident absolute prediction errors.

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Explainability analysis

Understanding why the model assigns specific competency scores is crucial for transparency and trust, particularly in clinical applications. SHAP (SHapley Additive exPlanations)[52] is a model-agnostic explainability method based on cooperative game theory that quantifies the contribution of each input feature to a specific prediction, ensuring consistent and additive attributions. Specifically, we adopted Antwarg et al.'s[53] Kernel SHAP approach for anomaly detection.

In our framework, MEMTO generates a segment-level anomaly score, which is then used as the scalar output to be explained. Following Antwarg et al.'s work, we treat MEMTO as a black-box scorer $f(x)$ and estimate per-variable Shapley values relative to a background set of 200 non-anomalous segments randomly drawn from the Rank 5 reference cohort. This anchors explanations to clinically meaningful patterns. For each sampled coalition, variables excluded from the coalition are masked channel-wise over the entire segment duration by substituting that channel with the corresponding channel from a randomly selected "ideal" segment, preserving the segment's temporal structure and cross-channel correlations; Kernel SHAP then fits the weighted local surrogate to obtain attributions. Positive SHAP values indicate variables that increase the anomaly score (depart further from expert patterns), whereas negative values

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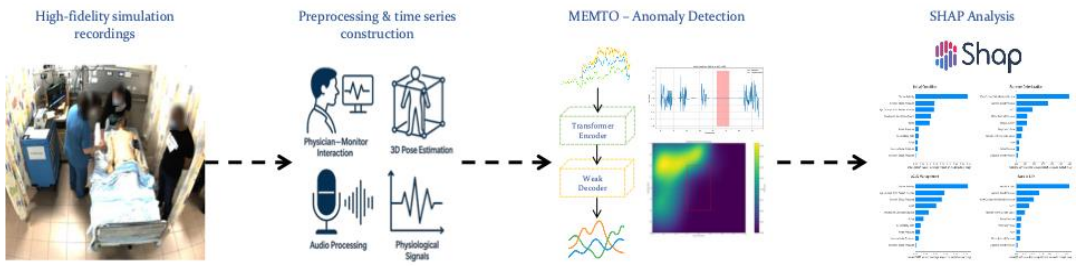
indicate variables that offset it. This procedure bridges MEMTO's quantitative anomalies with concrete, interpretable indicators without affecting model training or scoring. Our implementation follows Antwarg et al.'s Kernel SHAP setup; code and parameters are documented in the following repository: https://github.com/ronniemi/explainAnomaliesUsingSHAP.

Figure 4: Illustration of the overall pipeline. Multimodal feature extraction and temporal alignment, MEMTO-based anomaly scoring, aggregation and calibration to phase-specific 1–5 competency scores, and the subsequent SHAP-based explanation step.



Code availability

All requests for programming code should be addressed to SL. Any materials that can be shared will be released via a material transfer agreement for noncommercial research purposes.

All analyses were conducted on a computing cluster with two NVIDIA RTX A6000 48GB GPUs. The software environment consisted of Ubuntu 20.04 LTS, Python (v3.11), and PyTorch (v2.00).

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Data availability

All requests for raw and analyzed data, as well as related materials, will be reviewed by our legal department (Technion - Israel Institute of Technology) to verify whether the request is subject to any intellectual property or confidentiality constraints. Any data and materials that can be shared will be released via a material transfer agreement for noncommercial research purposes. The request should be addressed to SL.

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Authors contribution

SL was responsible for the conceptualization and overall supervision of the study, as well as securing the funding. SG led the formal analysis, methodology development, and visualization, and drafted the original manuscript. SG also performed validation and final editing under the guidance of SL and AR, who provided critical feedback and contributed to the review and editing process. FM conducted data curation and was responsible for data collection and project administration.
All authors reviewed and approved the final version of the manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

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